MICROSERVICES COURSE · SPRING BOOT & SPRING CLOUD

Professional Edition

Program Kickoff

Welcome · Roadmap · Platform · Working Model · Assessment

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ITSharks

IT Learning & Training

THE INDUSTRY CONTEXT

Why Microservices Matter Right Now

Netflix

Deploys to production
100s of times per day

Amazon

Deploys a new version
every 11.7 seconds

Uber

Runs 2,000+ microservices
in production

Industry leaders such as Netflix, Amazon, and Uber adopted microservices to achieve independent deployment, scalability, and team autonomy.

The ability to deploy independently, scale selectively, and own bounded domains is now a baseline expectation in any serious engineering organisation.

This course trains you to design, build, and operate systems the way industry leaders do.

WHAT CHANGES AFTER THIS COURSE

Problems This Course Equips You to Solve

Problem
A deploy in one module
breaks the entire system

Solution
Independent deployable services

Problem
One slow service
exhausts all threads

Solution
Resilience4j: CB + Bulkhead + TimeLimiter

Problem
Payment fails midway —
order is stuck in limbo

Solution
Saga Pattern + Compensation

Problem
8 services use
8 different configs

Solution
Spring Cloud Config Server

Problem
Order can't find
Inventory's address

Solution
Eureka Service Discovery

Problem
1000 requests/min
collapse the database

Solution
Redis Caching + Cache-Aside

FULL PROGRAM OVERVIEW

29 Sessions — 80 Hours — 4 Phases

Phase 1

Foundation & Core Patterns

Sessions 1–8 · 20h

- → Config + Eureka
- → API Gateway + Security
- → Resilience4j Stack
- → OpenFeign + Saga + Kafka
- → Redis Caching

Phase 2

Quality & Deployment

Sessions 9–16 · 20h

- → Docker + Compose
- → Integration Testing
- → Kubernetes
- → CI/CD Pipelines
- → GitOps + ArgoCD

Phase 3

Advanced & Enterprise

Sessions 17–24 · 20h

- → Observability Stack
- → CQRS + Event Sourcing
- → Keycloak + OAuth2
- → Service Mesh (Istio)
- → HashiCorp Vault

Phase 4

Capstone Project

Sessions 25–29 · 20h

- → Team Build Day
- → Production Deploy
- → Security Audit
- → Load Testing
- → Final Presentation

4 Offline Anchor Days (5h each) — Sessions 1, 9, 17, 25 · 25 Online Sessions (2.5h each) — Mon/Wed 3:00-5:30 PM

PHASE BY PHASE

From Foundation to Production — The Journey

Phase 1	Sessions 1–8 **Foundation & Core Patterns**	Config Server · Eureka · Gateway · JWT · Resilience4j · OpenFeign · Kafka Saga · Redis Cache
Phase 2	Sessions 9–16 **Quality & Deployment**	Docker · Integration Tests · Kubernetes · CI/CD · GitOps — the platform runs in production
Phase 3	Sessions 17–24 **Advanced & Enterprise**	Observability · CQRS · Security (Keycloak+Istio+Vault) · Performance — enterprise grade
Phase 4	Sessions 25–29 **Capstone**	Team build day · Full deployment · Security audit · Load test · Final viva presentation

THE CONTINUOUS CAPSTONE

You Build ONE Real Platform — Session 1 to Session 29

Enterprise E-Commerce Platform

A production-grade microservices system you will design, build, secure, test, and deploy.
Not a toy example. Not isolated exercises. One growing system — every session adds a real capability.

Every session has purpose

Your lab isn't a practice file — it's part of a real product you're building

Architecture decisions persist

A bad design in Session 3 costs you in Session 7 — just like real engineering

Ready for production

By Session 29, what you built can actually run in a real cloud environment

WHAT YOU WILL BUILD

Platform Architecture — Full Picture

API Gateway :8080

Product	Order	Payment	Inventory	Notification
:8081	:8082	:8083	:8084	:8085
S1	S4	S4	S6	S7

Infrastructure — running from Day 1, introduced when needed

Eureka (S1)	Config Server (S1)	PostgreSQL (S1)	Kafka (S7)	Redis (S8)	Zipkin (Later)
:8761	:8888	:5432	:9092	:6379	:9411

microservices-pro-platform · Branch: main (your work) + reference (instructor implementation)

WHY ONE PLATFORM, NOT MANY EXERCISES

The Continuous Capstone — Why It Matters

Traditional Training	Continuous Capstone (This Course)
New exercise every session — discarded after	One evolving platform — every lab builds on the last
Lab 3 has no relation to Lab 7	A design choice in Session 3 affects Session 7 — by design
"I built a circuit breaker demo"	"I added resilience to our Order Service"
Capstone starts from scratch in the final sessions	Capstone = completion of what you already built
Low motivation — throwaway work	High motivation — real product, real decisions

How the Course Is Delivered

Offline Anchor Day

4 Sessions (1, 9, 17, 25)

Wednesday · 10:00 AM – 3:00 PM

5 Hours

→ Deep-dive theory + whiteboard

→ Extended live coding (no rush)

→ Full supervised lab (90 min)

→ Architecture discussion + quiz

Online Live Session

25 Sessions

Monday & Wednesday · 3:00 – 5:30 PM

2.5 Hours

→ Knowledge Check on pre-reading

→ Concept + live coding (60 min)

→ Lab work — independent (35 min)

→ Quiz + Checkpoint Commit

EVERY SESSION — THE SAME PROVEN LOOP

How You Will Learn — The Session Loop

1 Pre-Reading

15 min before each session. Targeted material — not optional.

2 Knowledge Check

3 questions on the pre-reading, called by name at session start.

3 Concept + Problem

The problem is introduced first. Always. Then the solution.

4 Live Coding

Trainer codes live — you follow and type along. Real code, real mistakes.

5 Lab

You implement independently. 35 minutes of focused building.

6 Quiz

8 questions. 10 minutes. Immediate feedback on the session.

7 Checkpoint Commit

Commit and push your work. No commit = no grade for that lab.

↻ *This loop repeats for all 29 sessions*

HOW YOU WILL BE EVALUATED

Assessment Model — Four Components

Daily Quizzes

10%

8 questions · end of every session · 10 minutes · immediate feedback

Quick knowledge verification after every session

Lab Grades

40%

Feature (70%) + Unit Tests (20%) + Code Quality (10%) · one lab per session

The core grade: can you actually implement it?

Mid-Course Exam

15%

Session 9 · 2 hours · practical coding · individual · covers Sessions 1–8

Supervised exam at the Offline Anchor Day

Capstone Project

35%

Team 70% + Individual Contribution 30% · build, deploy, present, defend

The final product — running in production, presented as a team

Distinction: 90-100% **Pass with Merit:** 75-89% **Pass:** 60-74% **Retake Required:** < 60%

Expectations & Commitments

What We Expect From You

- **Pre-reading every session**
  15 min is enough — but it must be done
- **Labs completed between sessions**
  Don't wait for the next session to finish
- **Checkpoint commit pushed**
  No commit = no grade, no exceptions
- **Unit tests with every lab**
  Tests are not optional — 20% of your grade
- **Pre-reading before Knowledge Check**
  You will be called on by name, every session

What You Get From Us

- **Full Session Packs**
  Detailed instructor guide + code + labs for every session
- **Reference branch**
  Instructor implementation you can compare against
- **Architecture Clinics**
  Two full reviews (Sessions 8 & 24) of your design decisions
- **Assessment Bank**
  64 exam questions with explanations — all mapped to traps
- **Production-grade patterns**
  Patterns that work in real systems, not just demos

Kickoff Complete

Let's Start Session 1

Architecture & Spring Cloud · Service Discovery · Centralized Configuration

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